THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. If $(25)_b = 17$ in decimal (a two-digit number in base b), then b equals: 1

(a) 6 **(b)** 7 **(c)** 5 **(d)** 8

2. The angles of a quadrilateral are in the ratio 1:3:3:5. The difference between the largest and the smallest angle is: 1

(a) 150° **(b)** 132° **(c)** 120° **(d)** 180°

3. The smallest number which leaves remainder 7 when divided by 15 and 20 is: 1

(a) 127 **(b)** 307 **(c)** 60 **(d)** 67

4. $(a+b)^2 = a^2 + b^2$ is true only when: 1

(a) $a = b$ **(b)** $ab = 0$ **(c)** $a + b = 0$ **(d)** $a = 1$

5. The number x for which 4 : 9 = 12 : x is: 1

(a) 25 **(b)** 27 **(c)** 21 **(d)** 18

6. A sum triples in 4 years at simple interest. The rate of interest per annum is: 1

(a) 100% **(b)** 50% **(c)** 45% **(d)** 55%

7. The legs of a right triangle are in the ratio $8 : 15$ and its hypotenuse is 51 cm. Its perimeter (in cm) is: 1

(a) 102 **(b)** 69 **(c)** 120 **(d)** 123

8. The present ages of Ria and Sia are in the ratio 2:5. After 9 years the ratio will be 7:13. Ria's present age (years) is: 1

(a) 21 **(b)** 10 **(c)** 30 **(d)** 12

9. All the 16 faces of a polyhedron are pentagons. Its number of vertices is: 1

(a) 40 **(b)** 26 **(c)** 58 **(d)** 24

10. The mean of 5 numbers is 24. When one number is removed the mean of the remaining 4 becomes 21. The removed number is: 1

(a) 3 **(b)** 36 **(c)** 24 **(d)** 9

11. A two-digit number exceeds the number formed by reversing its digits by 36. The difference of its digits is: 1

(a) 36 **(b)** 5 **(c)** 4 **(d)** 3

12. The perimeter of a rectangle is 108 cm and the length is 5 times the breadth. Its area is: 1

(a) 729 cm² **(b)** 387 cm² **(c)** 405 cm² **(d)** 414 cm²

13. The number of perfect squares between 1000 and 3000 (both excluded) is: 1

(a) 22 **(b)** 24 **(c)** 25 **(d)** 23

14. 3^5 added 3 times equals: 1

(a) 9^5 **(b)** 3^{15} **(c)** 3^6 **(d)** 3^8

15. Weights of 1, 2, 4, 8, ... kg (powers of 2) are available. The least number of weights needed to weigh any whole number of kg up to 100 kg is: 1

(a) 6 **(b)** 8 **(c)** 7 **(d)** 10

16. One angle of a parallelogram is 3 times its adjacent angle. The smaller angle is: 1

(a) 135° **(b)** 60° **(c)** 65° **(d)** 45°

17. The number of multiples of 12 between 200 and 900 (both included) is: 1

(a) 58 **(b)** 60 **(c)** 59 **(d)** 61

18. If $x + y = 11$ and $xy = 28$, then $x^2 + y^2$ equals: 1

(a) 177 **(b)** 93 **(c)** 121 **(d)** 65

19. If x : y = 3 : 4 and y : z = 6 : 5, then x : y : z is: 1

(a) 18 : 24 : 20 **(b)** 3 : 4 : 5 **(c)** 9 : 12 : 10 **(d)** 3 : 10 : 5

20. Two articles are sold at ₹ 2400 each, one at a gain of 20% and the other at a loss of 20%. The overall result is a loss of (in %): 1

(a) 4% **(b)** 10% **(c)** 0% **(d)** 20%

21. The sides of a rectangle are in the ratio $5 : 12$ and its diagonal is 39 cm. The area of the rectangle (in cm²) is: 1

(a) 549 **(b)** 540 **(c)** 117 **(d)** 102

22. ₹ 3600 is divided among three friends in the ratio 1:3:8. The difference between the largest and the smallest share (in ₹) is: 1

(a) 900 **(b)** 2400 **(c)** 2100 **(d)** 300

23. A polyhedron has 30 edges and exactly 3 edges meet at every vertex. Its number of faces is: 1

(a) 32 **(b)** 20 **(c)** 10 **(d)** 12

24. The mean of 25 values was found to be 24. Later it was found that 90 was misread as 40. The correct mean is: 1

(a) 74 **(b)** 24 **(c)** 26 **(d)** 22

25. If $x + \frac{1}{x} = 4$, then $x^2 + \frac{1}{x^2}$ equals: 1

(a) 16 **(b)** 14 **(c)** 8 **(d)** 18

26. An isosceles trapezium has parallel sides 12 cm and 22 cm and non-parallel sides 13 cm. Its area is: 1

(a) 408 cm² **(b)** 204 cm² **(c)** 214 cm² **(d)** 221 cm²

27. The smallest number to be added to 2345 to make it a perfect square is: 1

(a) 55 **(b)** 56 **(c)** 57 **(d)** 58

28. If $3^{3x-2} = 3^{13}$, the value of x is: 1

(a) 15 **(b)** 6 **(c)** 3 **(d)** 5

29. Which of the following numbers is the smallest? 1

(a) 41_5 **(b)** 19_{10} **(c)** 10110_2 **(d)** 24_8

30. In a rhombus ABCD, $\angle ACD = 32^\circ$. Then $\angle ABC$ is: 1

(a) 64° **(b)** 116° **(c)** 148° **(d)** 58°

31. The number of 3-digit numbers divisible by 13 is: 1

(a) 70 **(b)** 69 **(c)** 76 **(d)** 68

32. If $x - y = 4$ and $xy = 45$, then $x^2 + y^2$ equals: 1

(a) 61 **(b)** -74 **(c)** 16 **(d)** 106

33. The present ages of two brothers are in the ratio 5 : 3. After 6 years the ratio becomes 3 : 2. The present age of the elder brother is: 1

(a) 30 years **(b)** 48 years **(c)** 18 years **(d)** 36 years

34. The population of a town is 40000 and it grows at 5% per annum. After 2 years it will be: 1

(a) 44000 **(b)** 48510 **(c)** 44100 **(d)** 39690

35. A Pythagorean triplet whose smallest member is 17 is: 1

(a) 17, 145, 146 **(b)** 17, 143, 144 **(c)** 17, 18, 19 **(d)** 17, 144, 145

36. In a pie chart a 75° sector represents 25 students. The total number of students is: 1

(a) 60 **(b)** 145 **(c)** 900 **(d)** 120

37. If every corner of a cube is sliced off, the number of faces of the resulting solid is: 1

(a) 6 **(b)** 12 **(c)** 8 **(d)** 14

38. A class has 24 students with mean age 12 years. When the teacher's age is included, the mean rises to 13. The teacher's age (years) is: 1

(a) 13 **(b)** 37 **(c)** 36 **(d)** 32

39. The sum of the dates in a 2×2 calendar block is 64. The smallest date in the block is: 1

(a) 20 **(b)** 13 **(c)** 12 **(d)** 11

40. The length of a rectangle is increased by 25% and its breadth decreased by 20%. If the original area is 2000 cm², the new area (in cm²) is: 1

(a) 2200 **(b)** 2500 **(c)** 2000 **(d)** 2100

41. **Assertion (A):** The ratio $15 : 25$ in its simplest form is $3 : 5$. 1

Reason (R): The ratio of two quantities of the same kind has no unit.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

42. **Assertion (A):** A shirt marked at ₹800 is sold at a 10% discount; its selling price is ₹720. 1

Reason (R): Profit or loss percent is calculated on the cost price.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

43. **Assertion (A):** $(5, 12, 13)$ is a Pythagorean triplet. 1

Reason (R): $(3, 4, 5)$ is a Pythagorean triplet.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

44. **Assertion (A):** For a fixed distance, the speed of a vehicle and the time taken are inversely proportional. 1

Reason (R): For a fixed time, the work done by workers is directly proportional to the number of workers.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

45. **Assertion (A):** A triangular prism has 5 faces. 1
Reason (R): A pyramid with an n-sided base has $n + 1$ faces.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** The median of 3, 7, 9, 12 and 15 is 9. 1
Reason (R): The mode of the data 2, 3, 3, 5 is 3.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** The expression $3x + 5$ has the value 17 when $x = 4$. 1
Reason (R): Terms with the same variable part are called like terms.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** The area of a triangle with base 10 cm and height 8 cm is 40 cm^2. 1
Reason (R): The area of a rectangle of length 10 cm and breadth 8 cm is 80 cm^2.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** The cube of a negative integer is negative. 1
Reason (R): The cube of every integer is positive.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** $3^0 + 2^0 = 2$. 1
Reason (R): $a^0 = 0$ for every non-zero number a.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**